

WASSWA WILSON

Biomedical Engineer · Researcher and Data Lead · Full-Stack Software Developer · Proposal Writer

Kampala, Uganda · wasswawilson0001@gmail.com · +256 783 230 321

English (fluent, professional working proficiency) · Luganda (native)

PROFESSIONAL PROFILE

I am a biomedical engineer, researcher and full-stack software developer based in Kampala, with over nine years of work across health research, data systems and software engineering. I started in research at the Uganda Virus Research Institute and I have kept the habit: I model a problem, test it against real data and document it so someone else can rely on it. During the COVID-19 response that produced the oxygen consumption formulas that gave hospital and national planners a defensible basis for sizing supply. Today I lead development of the FairBanks Community Health Intelligence Platform (FCHIP), where I own the data capture tools, the database, the cloud data flows and the management dashboards, and I am building a cross-sector application that turns field data into finished reports in real time. I also write the institutional proposals, grant applications and investment materials FairBanks submits. I analyse in Python, SQL, MATLAB and Excel, I build automated auditing tooling, and I design AI agents and automations that are running in production.

CAREER HIGHLIGHTS

- I developed the oxygen consumption formulas used to calculate patient demand during the COVID-19 response at International Hospital Kampala, and I supported Ministry of Health oxygen planning. I defined the model, tested it against real hospital demand and handed it to management as a planning tool. It went on to inform capacity planning for later facility projects.
- I write the proposals FairBanks submits, among them the WA Foundation Category 2 full project proposal for FairBanks Community Health Reach, accelerator and fellowship applications, and the investment proposals for the maternity and diagnostics projects. I handle the narrative, the financial annexes and the supporting evidence files.
- I lead development of FCHIP, the FairBanks Community Health Intelligence Platform. I built the community screening, referral and follow-up data capture, the database, the cloud sync and the reporting dashboards, and I deliver the platform across web and mobile.
- I am building a cross-sector data collection and reporting application that writes the report as the user works, instead of leaving it to be compiled afterwards. I designed it for accounting, research, biology, agriculture, social media and similar fields.
- I designed, built and deployed an AI reception agent for a law firm in Texas, USA. It handles live inbound client calls, books and reschedules appointments, and runs client follow-up. It is in active production use.
- I prepared and maintained the biomedical compliance evidence that carried International Hospital Kampala through successful COHSASA accreditation, which is data quality assurance against a published standard.
- I have delivered hospital technology projects in five countries: Uganda, the Democratic Republic of Congo, Kenya, Tanzania and Somaliland, and I upgraded partner facilities for the Gould Family Foundation including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi.

RESEARCH, DATA AND PROPOSAL DEVELOPMENT

Research

I worked as a research intern at the Uganda Virus Research Institute on laboratory data analysis, mathematical modelling and research pipeline design in Python. My BSc at Makerere covered Research Methods, Biomedical Modelling and Simulation, Biosignal Processing and Analysis, and Bioinformatics and Functional Genomics, and I carried a full research project through to completion. I apply the same method to operational problems, most visibly in the COVID-19 oxygen demand modelling.

Data management at scale

I design and run the data layer of FCHIP: capture schemas, validation rules, cloud data flows, storage and synchronisation across web and mobile. Earlier I built and maintained the equipment registers, maintenance schedules and service histories for multi-facility programmes at International Hospital Kampala and the Gould Family Foundation, and kept them accurate enough to survive accreditation review.

Analysis and reporting

I analyse in Python, SQL, MATLAB and Excel. I publish results as live management dashboards rather than static documents, so decision-makers work from current numbers, and I build automated auditing and verification tooling that checks records in bulk instead of by hand.

Proposals, grants and technical writing

I write and edit institutional proposals, grant applications, investment proposals, concept notes, partnership documents, pitch decks, technical documentation and user manuals, together with the annexes and evidence files that have to stand up to assessment.

Modern tooling and automation

I work with AI agents and agent-assisted development daily, in Cursor, Claude, OpenAI GPT and Codex, and I build automation pipelines that take routine collection, reconciliation and reporting off people's hands.

TECHNICAL SKILLS

Data, analytics and reporting

Python · SQL · MATLAB · Excel · relational modelling and query optimisation · data cleaning and validation · ETL and cloud data pipelines · dashboard and report design · automated auditing and verification tooling

AI, agents and automation

AI agent design, deployment and monitoring · workflow and process automation · third-party systems integration · Cursor · Claude · OpenAI GPT · Codex

Programming languages

Java · Python · JavaScript · TypeScript · C · C++ · C# · PHP · SQL · MATLAB · Arduino C · HTML5 · CSS3

Web and mobile

React · Next.js (App Router) · React Native · Expo · Tailwind CSS · styled-components · Bootstrap · JavaFX / FXML · Jest · Playwright

Backend and APIs

Node.js · Express · Laravel · CodeIgniter · Spring / Spring Boot · Convex · Prisma · MySQL · REST APIs · JWT · role-based access control

Cloud and DevOps

AWS · Microsoft Azure · Vercel · EAS · Bunny.net CDN · Git and GitHub · CI/CD pipelines · release and environment management

Research and scientific

Research methods and study documentation · mathematical modelling and simulation · biosignal processing · medical imaging · bioinformatics and functional genomics · embedded systems design · Proteus · Eagle CAD · Solid Edge

Documents and productivity

Microsoft Word · Excel · PowerPoint · proposal, grant and investment writing · technical writing · document editing and design · Windows, Linux and Android

Medical equipment

CT and X-ray imaging · ultrasound and C-arms · ventilators, monitors and infusion pumps · theatre and ICU systems · haematology, chemistry and blood bank analysers · neonatal incubators, phototherapy, foetal Dopplers and CTG · oxygen plants, cylinder and manifold systems and piped medical gas · RO water and sterilisation plant

PROFESSIONAL EXPERIENCE

Lead Software Developer, FCHIP and Health Information Systems

2026 to present

FairBanks Medical Centre Limited, Kampala, Uganda

- I lead the design and build of the FairBanks Community Health Intelligence Platform (FCHIP). I do the data capture form design, the database structure, the cloud data flows, the reporting dashboards and the delivery across both web and mobile.
- I define the indicators, validation rules and data-quality checks behind FCHIP reporting, and I analyse the resulting data in Python and SQL so that management reporting rests on figures that have been checked.
- I write and edit the institutional proposals, grant applications, investment materials and partnership documents FairBanks submits, including the WA Foundation Category 2 full project proposal for FairBanks Community Health Reach. I chaired the FairBanks meeting with the Uganda Health Partners Cooperative on cooperative health insurance in July 2026.
- I am building a cross-sector data collection and reporting application that turns work already being done into finished reports in real time, for accounting, research and laboratory science, biology, agriculture, social media and similar record-heavy fields.
- I design and deploy the AI agents and automation pipelines that remove manual handling from reception, appointment management, client follow-up and routine reporting, and I architect and maintain the hospital and clinic software that staff use daily.
- I own the technical documentation, the data-protection practice, the access control and the release management across the platform.

Consultant Biomedical Engineer, Independent Practice

Mar 2025 to present

Contract assignments: Uganda, DR Congo, Kenya, Tanzania, Somaliland

- I install, commission, calibrate and maintain hospital medical equipment under contract across five countries, working independently on site and to a fixed handover standard.
- I completed the laboratory, theatre and radiology installations at several hospitals in the Democratic Republic of Congo, including the facility serving the Kibali Gold Mine.
- I train the clinical users at handover, and I produce a written service report and maintenance record for every assignment, so the client keeps a verifiable equipment history.

Biomedical Programs Manager

Aug 2024 to Feb 2025

Gould Family Foundation (GFF), East and Central Africa

- I managed the regional biomedical team, covering programme planning, implementation and reporting across partner health facilities in multiple countries.
- I built and maintained the programme data: equipment registers, lifecycle records, procurement documentation and the routine reports that went to management.
- I developed the equipment lifecycle-management and master-planning strategies that were then applied across partner facilities.
- I upgraded several facilities, including the neonatal intensive care unit at Mama Lucy Kibaki Hospital, Nairobi, where my oxygen and equipment demand modelling informed the capacity plan.
- I led the biomedical and clinical user training programmes for engineers, technicians and clinical staff.

Biomedical Manager

Jan 2020 to Jan 2024

International Hospital Kampala (IHK)

- I managed the biomedical engineering function of a major private hospital for four years, including team supervision, workplans and reporting to executive management.
- I led oxygen therapy management through the COVID-19 response and I supported Ministry of Health oxygen planning. I developed the formulas used to calculate patient oxygen consumption, which gave managers a dependable basis for sizing plant capacity, cylinder stock and flow requirements rather than estimating them.
- I prepared and maintained the compliance evidence files reviewed for COHSASA accreditation, and I supported the hospital through successful accreditation.
- I led the installation and commissioning of CT and X-ray imaging, oxygen plant, laboratory, theatre and ICU systems through to handover, and I trained the clinical users.
- I built the preventive maintenance schedules and service-record systems that reduced equipment downtime, and I coordinated procurement, technical specification and lifecycle management of hospital medical equipment.

Biomedical Engineer

Apr 2019 to Jan 2020

Norvik Hospital Ltd, Kampala

- I maintained and serviced the hospital diagnostic and patient-monitoring equipment, and I supported the installation and calibration of imaging and ICU equipment.
- I helped establish the preventive maintenance schedules and service follow-up records.

Research Intern

Dec 2018 to Apr 2019

Uganda Virus Research Institute (UVRI)

- I carried out laboratory data analysis, mathematical modelling and research pipeline design in Python.
- I contributed to biomedical research projects and laboratory experimentation, and I handled the recording, quality checking and documentation of research data so results could be reproduced.

Teaching Assistant

Mar 2016 to Aug 2017

Makerere University, College of Health Sciences

- I assisted in teaching biomedical engineering course units at undergraduate level.
- I supervised and supported the laboratory practical sessions for engineering students.

SELECTED PROJECTS

Institutional proposals, grants and investment materials

2026 to present

Lead Writer · FairBanks Medical Centre Limited

I write and edit what FairBanks submits: the WA Foundation Category 2 full project proposal for FairBanks Community Health Reach, accelerator and fellowship applications, investment proposals for the maternity and diagnostics projects, partnership documents and pitch decks. I handle the narrative, the annexes, the financial presentation and the evidence files, and I maintain the pipeline of opportunities and deadlines.

COVID-19 oxygen therapy management and demand modelling

2020 to 2022

Biomedical Manager · International Hospital Kampala, with Ministry of Health planning support

I ran oxygen therapy management for the hospital through the pandemic and I developed the formulas for calculating patient oxygen consumption. My figures were used to size plant capacity, cylinder stock and flow requirements, and to support planning decisions at managerial and national level.

AI reception agent

2025 to present

Designer and Developer · Law firm, Texas, USA

I built and deployed an AI agent that answers live inbound client calls, books and reschedules appointments, and runs post-contact client follow-up. It is in active production use.

Shulekeeper school information system

2021 to 2024

Assistant Software Developer · Shulekeeper

I contributed front-end and back-end development, third-party integrations, testing, user training and documentation to a multi-user records and reporting platform built with React and Next.js, TypeScript, Laravel, MySQL and AWS.

Endoscopy image capture and reporting software

2017 to 2019

Lead Software Developer · St. Catherine Hospital and St. Francis Hospital

I led the build of a clinical image capture and procedure reporting system in JavaFX and MySQL. I did the algorithm design, the relational modelling, the legacy data migration, the on-site installation, the user training and the user manual.

Wekebere foetal heart-rate monitoring system

2017 to 2024

Electronics Developer and Programmer · Wekebere

I built an Android and Arduino monitoring system for third-trimester foetal heart-rate measurement, and I implemented the Java serial communication between device and phone, with Microsoft Azure cloud storage.

EDUCATION

Certificate in Leadership and Management in Health

2022

University of Washington

Grade A+. Planning, leadership and management of health programmes.

BSc Biomedical Engineering, Second Upper Honours

2012 to 2017

Makerere University, Kampala

Research-relevant course units: Research Methods, Biomedical Modelling and Simulation, Biosignal Processing and Analysis, Bioinformatics and Functional Genomics, Database Systems and Software Engineering. I completed a full research project as part of the degree.

Advanced Java Programming, Level 1

2015 to 2016

Green Bridge School of Open Technologies

Grade A+.

Uganda Advanced Certificate of Education (UACE)

2011

Mengo Secondary School

Principal subjects: Mathematics, Physics, Biology, Chemistry.

Uganda Certificate of Education (UCE)

2009

St. John's Wakiso Secondary School

Division 1.

TRAINING AND CERTIFICATION

- **COHSASA accreditation standards: in-service preparation and compliance.** International Hospital Kampala, 2020 to 2024.
- **Fire Safety, Prevention, Firefighting and Emergency Scene Management.** Fire Technologies Limited (IHK), 2021.

INTERESTS

I play music, the saxophone in particular. · I watch film. · I train and exercise regularly. · I enjoy travelling and working in unfamiliar places, which the regional field work has made second nature.

REFEREES

Eng. Richard Ssejongo

Biomedical Engineer, St. Francis
Hospital Nsambya

+256 753 818 754 · +256 777 132 489

Dr. Annet Khingi

Senior Radiologist and Administrator,
Mengo Hospital

+256 772 592 771 · +256 701 592 771

Racheal Nabukeera

Director and Founder, FairBanks
Medical Centre

+256 772 849 258 ·
nracheal017@gmail.com